A Phased Workflow for Operating LLM-Based Coding Agents

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Agents write code faster
than humans can review it

- Losing track of the agent costs. A 25% rise in AI adoption came with an estimated 7.2% drop in delivery stability [1]. One agent deleted a production database during a code freeze [2].
- Watching is not enough. Operators who only monitor lose the skill to take over [3].
- Autopilot kept the pilot, who works hardest at takeoff and landing. Our workflow puts human effort in the same places.

Long, noisy context makes agents worse

Multi-turn chats cut LLM performance by **39%** on average [4]. Irrelevant context lowers accuracy even when the answer is present [5].

Failure modes [6]

distraction	old patterns dominate
confusion	irrelevant content
poisoning	hallucinations persist
clash	new facts conflict

Strategies against them [7]

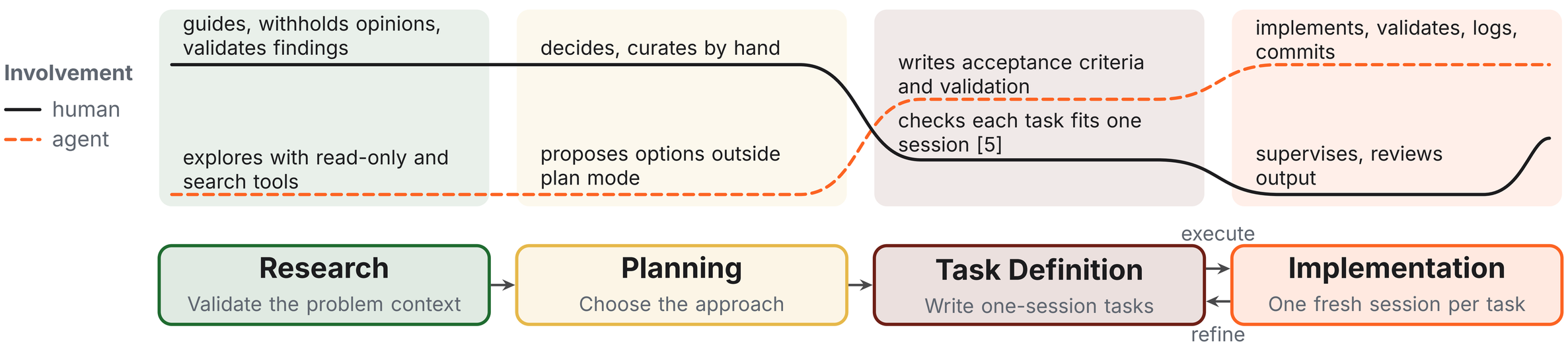
write	save to files
select	load what's needed
compress	keep what's required
isolate	fresh sessions

Front-load the thinking.
Humans own research and planning, where mistakes cost the most.

Build the backpressure.
Acceptance criteria and validation reject bad output before you review it.

Stay on the loop.
One task per fresh session. You review, and bad tasks go back.

Four phases, with human effort front-loaded

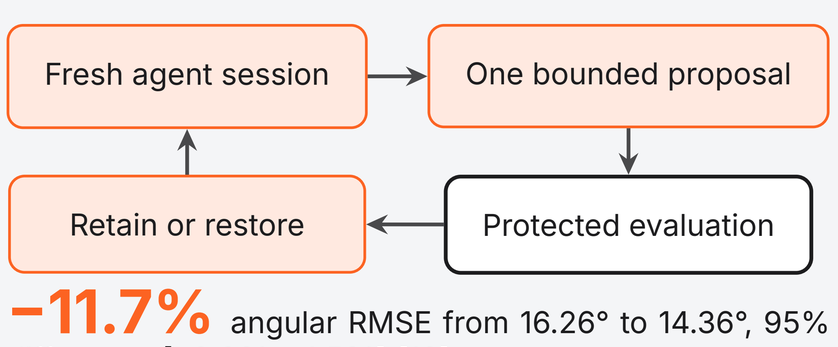


	Research	Planning	Task Definition	Implementation
Prevents	*confusion* *distraction*	*poisoning*	*distraction*	*poisoning* *distraction*
Using	**select** **isolate** **write**	**compress** **write**	**write** **isolate**	**isolate** **select** **write**
Repo skills	`/research` `/scholar` `/handover`	`/brainstorm` `/handover`	`/to-tasks`	manual supervision, autonomous loop
Writes	`RESEARCH.md`	`PLAN.md`	`TASKS.md`	`AGENTS.md` `ACTIVITY.md` `git commits`

Research and planning run once per project. Each repeated agent mistake earns one short line in `AGENTS.md` [8]. Extends research-plan-implement [9].

Open-ended research, beyond the paper

An autonomous implementation loop improved PCA normal estimation [10] on PCPNet [11]. The agent could edit the estimator but never the scorer.



322 fresh sessions

<18 h wall clock

−11.7% angular RMSE from 16.26° to 14.36°, 95% CI of the difference [−2.23°, −1.59°] [12]

No single session found the final estimator. With no control run, the gain is not yet the workflow's (open problem 1).

Let's talk about open problems

1. **No metrics for workflow effectiveness.** Nobody can yet show that a phased workflow beats unstructured delegation, or that tasks have the right size.
   Candidate metrics, none tested yet
   - *task success rate per session*
   - *instruction correction rate*
   - *context utilization at task completion*
2. **Research stops at components.** Pruning, compression, and tool selection are studied one at a time. The workflows that combine them have no shared terms, so teams cannot compare them.

Have a metric or a counterexample? Open an issue in the repo.

infobip/agentic-development-workflow

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